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Системная эпигенетическая дисрегуляция как фактор старения и терапевтическая мишень

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Абстрактный

Несмотря на то, что эпигенетические изменения в процессе старения хорошо изучены, у нас нет единой системы для систематического объяснения их механизмов. В этом обзоре мы представляем концепцию системного уровня, которая демонстрирует, как эпигенетическая регуляция влияет на процесс старения. Мы рассматриваем четыре взаимозависимых процесса, из-за которых эпигенетическая точность — способность систем регуляции хроматина поддерживать точное состояние экспрессии генов — постепенно снижается: ухудшение ядерной архитектуры, в том числе разрушение доменов, ассоциированных с ламиной; нарушение регуляции эпигенетической памяти с помощью комплексов, модифицирующих хроматин, таких как репрессивный комплекс 2 группы Polycomb (PRC2); изменения нуклеосом, связанные с независимым от репликации накоплением варианта гистона H3.3; и перепрограммирование транскрипции под действием факторов транскрипции. Эти процессы взаимодействуют посредством перекрестной регуляторной обратной связи, что приводит к каскадным сбоям в экспрессии генов и поддержании клеточного состояния. Эта концепция

объясняет, почему терапевтические методы, нацеленные на эпигенетические системы, дают стабильный эффект в различных модельных системах и при различных фенотипах старения. Взаимосвязанная организация механизмов регуляции хроматина создает конкретные терапевтические цели для восстановления регуляторной согласованности. Раскрывая механизмы, с помощью которых эпигенетическая дисрегуляция приводит к фенотипам старения, мы стремимся обеспечить рациональный подход к разработке терапевтических средств, нацеленных на эпигенетические системы, которые дают сбой при старении, а не на отдельные молекулярные дефекты.

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Cellular Senescence and Age-Associated Mechanisms

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Competing interests

The authors declare no competing interests.

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Supplementary information

Supplementary Information (download PDF ↓)

Glossary

Bistability

Co-existence in a chromatin region of two stable epigenetic states, with transitions between states requiring crossing an energy barrier.

Bivalent

Refers to chromatin domains that simultaneously carry gene-activating and gene-repressing histone modifications.

Enhancer–promoter loops

Physical interactions between distant enhancer elements and gene promoters that enable gene activation.

Epigenetic drift

The progressive accumulation of both random and directional changes in chromatin states during ageing that disrupts gene regulation.

Epigenetic memory

The maintenance of gene expression states through cell divisions without changes in DNA sequence.

Intermediate genomic compartment

A compartment formed at the interface of genomic A and B compartments and that holds an intermediate spatial position.

Lamina-associated domains

(LADs). Large chromatin regions associated with the nuclear lamina that are typically gene-poor and transcriptionally repressed.

Nuclear lamina

The protein meshwork lining the inner nuclear membrane; supports nuclear structure and helps organize chromatin.

Pioneer transcription factors

Transcription factors capable of binding to condensed chromatin and initiating its opening.

Polycomb repressive complex 2

(PRC2). A chromatin modifier that trimethylates histone H3 Lys27 (H3K27me3) to maintain gene silencing and cellular identity.

Positional memory

The capacity of chromatin-modifying complexes to maintain their genomic localization following cell division, typically through self-reinforcing biochemical feedback such as recognition of their own catalytic products.

Radial disorder

Increased cell-to-cell variability in the radial positioning of genomic loci within the nucleus.

Recruitment logic

The rules governing how chromatin-modifying complexes are recruited to specific genomic loci, including DNA sequence motifs, pre-existing histone modifications, transcription factor interactions and noncoding RNA binding.

Senescence-associated secretory phenotype

(SASP). The secretion of inflammatory cytokines and other factors by senescent cells.

Shannon entropy

An information-theory measure of disorder in a distribution.

Substrate drift

In this context, age-associated changes in histone variant landscapes.

Topologically associating domains

(TADs). Three-dimensional chromatin domains that group genes with their regulatory elements and insulate them from neighbouring regions.

Waddington landscape

A metaphor for how gene regulation modulates cell fate, visualizing development as marbles rolling down a landscape of valleys.

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